

CSL253 - Theory of Computation

Tutorial 7

Team Members

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Question 6

Let $\text{INFINITEDFA} = \{\langle A \rangle \mid A \text{ is a DFA and } L(A) \text{ is an infinite language}\}$. Show that INFINITEDFA is decidable.

Problem 6: Decidability of $\text{INFINITE}_{\text{DFA}}$

Show that $\text{INFINITE}_{\text{DFA}}$ is decidable.

On input $\langle A \rangle$, where A is a DFA:

1. Let k be the number of states of A .
2. Construct a DFA D that accepts all strings of length $\geq k$.
3. Construct a DFA M such that

$$L(M) = L(A) \cap L(D)$$

where:

- $L(A)$: the original language accepted by DFA A ,
- $L(D)$: the set of all strings of length $\geq k$, where k is the number of states in A .

So, $L(M)$ means the set of all strings that:

- Are accepted by A , and
- Have length $\geq k$.

- This implies: if the intersection is not $\emptyset$, then there exists a string of length $\geq k$. But since A has k states, according to the Pumping Lemma, there must be a loop in the DFA.
- Therefore, we can pump (i.e., repeat) that loop any number of times to generate strings of arbitrary length, meaning A accepts an infinite number of strings.

4. Test whether $L(M) = \emptyset$ using the EDFA decider T :

- If T says $L(M) = \emptyset$ (i.e., no string of length $\geq k$ is accepted), then A is finite, so **reject**.
- Otherwise, A accepts infinitely many strings (since some strings of length $\geq k$ are accepted), so **accept**.

$\Rightarrow$ If T accepts, reject; if T rejects, accept.

EDFA is a decidable language

Decider T :

On input $\langle A \rangle$, where A is a DFA:

1. Mark the start state q_0 .
2. Repeat until no new states get marked:
 - Explore all states that can be reached from the start state.
3. Mark any state that has an incoming transition from an already marked state:
 - If a state is reachable from a marked state, we also mark it.
4. Final check:
 - If no accept state is marked, it means no accepting state is reachable. So $L(A) = \emptyset \Rightarrow$ **accept**.
 - Otherwise, if at least one accept state is marked, then A accepts some strings. So $L(A) \neq \emptyset \Rightarrow$ **reject**.